



Member of GHM GROUP



## MEASURING **SOLAR** POWER IN THE MOST EFFECTIVE AND ACCURATE WAY

Buy **experience:**  
Delta OHM supplies Pyranometers to the market since over 20 years

Buy **efficiency:**  
our Pyranometers are easy to set up and quick to install

Buy **accuracy:**  
we supply each Pyranometer with an individual calibration report

Buy **the best:**  
our Pyranometers are the only ones on the market with 6 years' warranty



| Technical Specifications                                                 |                                                         | LPPYRA10                                        | LPPYRA02                                        | LPPYRA03                                        |
|--------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| WMO/ISO Classification                                                   |                                                         | Secondary standard                              | First Class                                     | Second Class                                    |
| ISO 9060:2018*                                                           |                                                         | Spectrally flat Class A (+++)                   | Spectrally flat Class B (++)                    | Spectrally flat Class C (+)                     |
| Typical sensitivity                                                      |                                                         | 6 to 11 $\mu\text{V}/(\text{W}/\text{m}^2)$     | 6 to 12 $\mu\text{V}/(\text{W}/\text{m}^2)$     | 50 to 15 $\mu\text{V}/(\text{W}/\text{m}^2)$    |
| Impedance                                                                |                                                         | 5 $\Omega \div 50 \Omega$                       | 33 $\Omega \div 45 \Omega$                      | 33 $\Omega \div 45 \Omega$                      |
| Measuring range                                                          |                                                         | 0-4000 $\text{W}/\text{m}^2$                    | 0 $\div$ 2000 $\text{W}/\text{m}^2$             | 0 $\div$ 2000 $\text{W}/\text{m}^2$             |
| Viewing field                                                            |                                                         | 2 $\pi$ sr                                      | 2 $\pi$ sr                                      | 2 $\pi$ sr                                      |
| Spectral range (50%)                                                     |                                                         | 283 nm ... 2800 nm                              | 283nm $\div$ 2800 nm                            | 300nm $\div$ 2800 nm                            |
| Operating and storage temperature range                                  |                                                         | -40 $^{\circ}\text{C} \div 80 ^{\circ}\text{C}$ | -40 $^{\circ}\text{C} \div 80 ^{\circ}\text{C}$ | -40 $^{\circ}\text{C} \div 80 ^{\circ}\text{C}$ |
| Weight                                                                   |                                                         | 0.90 Kg                                         | 0.90 Kg                                         | 0.45 KG                                         |
| ISO 9060 Specifications                                                  |                                                         |                                                 |                                                 |                                                 |
| Response time 95%                                                        |                                                         | < 5 sec.                                        | < 28 sec                                        | < 30sec                                         |
| Zero Off-set                                                             | a) Response to thermal radiation (200Wm <sup>-2</sup> ) | <7 $\text{W}/\text{m}^2$                        | 15 $\text{W}/\text{m}^2$                        | 25 $\text{W}/\text{m}^2$                        |
|                                                                          | b) Response to temperature change 5K/h                  | <  $\pm 2$   $\text{W}/\text{m}^2$              | <  $\pm 4$   $\text{W}/\text{m}^2$              | <  $\pm 6$   $\text{W}/\text{m}^2$              |
| Non stability over 1 year                                                |                                                         | <  $\pm 0.5$   %                                | <  $\pm 1.5$   %                                | <  $\pm 2.5$   %                                |
| Non-linearity                                                            |                                                         | <  $\pm 0.2$   %                                | <  $\pm 1$   %                                  | <  $\pm 2$   %                                  |
| Cosine response                                                          |                                                         | <   $\div 10$   $\text{W}/\text{m}^2$           | <  $\pm 18$   $\text{W}/\text{m}^2$             | <  $\pm 22$   $\text{W}/\text{m}^2$             |
| Spectral selectivity                                                     |                                                         | <   $\pm 3$   %                                 | <  $\pm 5$   %                                  | <  $\pm 7$   %                                  |
| Temperature response (-10 $^{\circ}\text{C}$ to +40 $^{\circ}\text{C}$ ) |                                                         | <1 %                                            | < 4 %                                           | < 8 %                                           |
| Tilt response                                                            |                                                         | <  $\pm 0.2$   %                                | <  $\pm 2$   %                                  | <  $\pm 4$   %                                  |
| Humidity Range                                                           |                                                         | 0 to 100%                                       | 0 to 100%                                       | 0 to 100%                                       |
| Mean Time Between Failures                                               |                                                         | > 10 years                                      | > 10 years                                      | > 10 years                                      |
| Accuracy of bubble level                                                 |                                                         | < 0.1 $^{\circ}$                                | <0.1 $^{\circ}$                                 | <0.2 $^{\circ}$                                 |
| Ingress Protection (IP) rating                                           |                                                         | 67                                              | 67                                              | 67                                              |
| Version with shadow ring                                                 |                                                         | ✓                                               | ✓                                               | ✓<br>(only for passive version)                 |
| Warrenty                                                                 |                                                         | 6<br>(registration required)                    | 6<br>(registration required)                    | 3                                               |
| ISO 17025 Certification                                                  |                                                         | on request                                      | on request                                      | on request                                      |

\*Under development. Release planned for September2018



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The qualitative level of our instruments is the result of a continuous evolving of the product itself. We reserve the right to perform modifications and corrections at any time without notice.

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